

“EA Spouse” and the Crisis of Video Game Labour: Enjoyment, Exclusion, Exploitation, Exodus

Nick Dyer-Witthford
University of Western Ontario

Greig de Peuter
Simon Fraser University

Abstract: The blog postings of “EA Spouse,” partner of an exhausted video game programmer, have catalyzed discussion of epidemic overwork in the digital play industry. This paper analyzes the crisis of labour in this glamorous new medium. After a brief overview of the industry and its production process, we discuss its labour conditions under four headings. “Enjoyment” examines the real pleasures game workers find at their jobs. “Exclusion” discusses the gendering of game work. “Exploitation” investigates the corporate processes that drive toward a work culture of extreme hours and the consequences game workers suffer. “Exodus” looks at current attempts by workers to escape this predicament— attempts including legal action, educational efforts, entrepreneurial flight, and union organizing.

Resume: Le blogue de « EA Spouse », conjointe d’un programmeur de jeux vidéo épuisé professionnellement, a inspiré bon nombre de discussions sur le surmenage qui sévit dans l’industrie du jeu vidéo. Cet article analyse la crise de la main-d’œuvre dans ce nouveau secteur glamour. Suivant un bref exposé général de l’industrie et de son mode de production, nous discutons en quatre étapes les conditions de travail qui y règnent. « Plaisir » porte sur les vrais plaisirs qu’éprouvent les travailleurs dans ce secteur. « Exclusion » porte sur la proportion trop faible de femmes qui y œuvrent. « Exploitation » explore la culture organisationnelle qui contraint les programmeurs à travailler pendant de très longues heures et les conséquences du surmenage sur ceux-ci. « Exode » observe comment les travailleurs tentent actuellement d’échapper à cette situation. Ces tentatives comprennent l’action judiciaire, l’éducation des travailleurs, l’exode des entrepreneurs et l’organisation de syndicats.

Keywords: Video game industry; Political economy

Nick Dyer-Witthford is Associate Professor in the Faculty of Information and Media Studies at the University of Western Ontario, London, ON N6A 5B7. E-mail: ncdyerwi@uwo.ca. Greig de Peuter is a doctoral candidate in the School of Communication at Simon Fraser University, Burnaby, BC V5A 1S6. E-mail: gsdepeut@sfu.ca.

Canadian Journal of Communication, Vol 31 (2006) 599-617

©2006 Canadian Journal of Communication Corporation

Introduction: "My happy supportive smile is running out"

On November 10, 2004, an open letter posted to a blog titled "EA: The Human Story" sent shockwaves through one of the coolest sites of contemporary media labour. Signed by "EA Spouse," the letter was authored by the "significant other" of an employee of one of the video game industry's largest corporations, Electronic Arts (EA). EA Spouse (2004) described how initial enthusiasm for a job with a company listed as one of Fortune's "100 Best Companies to Work For" had evaporated, as seven-day, 85-hour work weeks, uncompensated either by overtime pay or time off, became routine. It told of EA's "put up or shut up and leave . . . human resources policy"; of its dubious invocation of California labour-law exemptions on "specialty" employees to avoid paying overtime; and of creativity decomposing in a "money farm" churning out commercially safe game designs. Describing an industry pressing its workers "to individual physical health limits," EA Spouse wrote of how "the love of my life is coming home late at night complaining of a headache that will not go away and a chronically upset stomach, and my happy supportive smile is running out." She concluded with a question for EA's CEO:

[Y]ou do realize what you're doing to your people, right? . . . That when you keep our husbands and wives and children in the office for ninety hours a week, sending them home exhausted and numb and frustrated with their lives, it's not just them you're hurting, but everyone around them, everyone who loves them? When you make your profit calculations and your cost analysis, you know that a great measure of that cost is being paid in raw human dignity, right? (EA Spouse, 2004)

As responses to the post poured in, and as countless websites linked to the letter, it rapidly became obvious that EA Spouse's narrative, far from being an isolated case, articulated a seething reservoir of resentment and discontent within the video game industry.¹

This paper examines the conditions that generated the crisis in video game labour exposed by EA Spouse and the variety of responses this exposure has elicited from both game corporations and game workers. This empirically grounded case study is in part intended as a contribution to the theoretical discourses surrounding the concept of immaterial labour, a concept that has received considerable attention (e.g., Terranova, 2004; Wright, 2006) in the wake of the widely circulated writings on the topic by Michael Hardt & Antonio Negri in *Empire* and *Multitude* (2000; 2004). This literature lies largely outside the academic borders of communication studies but is nonetheless complementary to the growing body of contemporary scholarship in political economy and cultural studies on the conditions of media and cultural labour (e.g., McRobbie, 2002; Menzies, 2005; Ross, 2003). Immaterial labour has been defined broadly by Hardt & Negri as that which "involves communication, cooperation, and the production and reproduction of affects" (2000, p. 53). This spans technological work with computers and networks, work manipulating emotion—generating a sense of ease or excitement—and work involving communication and coordination. Discussions of immaterial labour have usefully identified some of the core features and tendencies of

contemporary capitalist production under advanced post-Fordism. However, this term is frequently used abstractly, at some disconnect from the experiences, hierarchies, and oppressions of such labour on the ground, in bodies, within specific sectors (cf. Tsianos & Papadopoulos, 2006). To counter the sometimes disembodied terminology deployed in the discourse surrounding immaterial labour (Dyer-Witheford, 2005), we offer the following analytical portrait of the composition of one particular strata of immaterial labour. Such grounded inquiry is the foundation from which the theorizing of communicative labour must proceed.

Our analysis of the preconditions and consequences of EA Spouse's intervention confirms that the game workplace is a site of conflict, rather than of acquiescence. The struggle between game capital and game labour therein is critically analyzed under four headings: "Enjoyment," "Exclusion," "Exploitation," and "Exodus." These, we argue, are points on a continuum of control, discontent, fight, and flight. For many, the initially enjoyable aspects of work in digital play mutate into a linchpin of exploitative and exclusionary practices. In terms of exclusion, we note that game development studios are deeply gendered, a structure of inequality in which, we suggest, the (excessive) work routines to which a largely male-dominated work force are subjected are sustained by a largely female-conducted sphere of invisible, unpaid caring labour. Of the various manifestations of exploitation in this industry, we concentrate in these pages on just one: overwork. After the EA Spouse episode, game studios cannot hide their endemically excessive hours—in turn a major source of game labour's fights and flights. Those game workers who are beginning to speak out are revealing the rust on the "digital sublime" mythology (Mosco, 2004); in the process, they are not only exposing the false promise of being paid to play, but also broaching the topic of labour organizing, a development whose significance, we believe, cannot be overstated. For the game industry is historically steeped in the ludic entrepreneurialism of the "Californian ideology" (Barbrook & Cameron, 1996) and thus presumed to be fundamentally at odds with the definition of the workplace as a site of labour conflict, be it through class-action lawsuits or the traditions of trade unionism.

Although we focus in this paper on the ostensibly glamorous technical and artistic labour involved in game development in the cities of the global North, it must not be forgotten that these immaterial labourers remain extraordinarily privileged in terms of the planetary hierarchy of labour. The game industry, like other digital businesses, is dependent on a manufacturing work force located in maquiladoras and free-enterprise zones in Central and South America, Eastern Europe, Southeast Asia, and China (de Peuter & Dyer-Witheford, 2005; Dyer-Witheford, 1999, 2001; Lugo & Lossada, 2002). At the depths of gaming's hidden abodes of production lie links to activities such as the mining of columbine tantalite (a rare mineral used in game consoles and other high-tech commodities) in the Congo and Egypt. The dependence of digital play on this highly exploited assembly and extraction work must be remembered, lest we fall right back into the above-mentioned problem of ethereality—and, in so doing, forget that it is not just glam-

orous, but also deadly, labour that makes digital play possible. This is not to say, however, that the conditions of these immaterial game labourers are unimportant. Indeed, as we shall see, there is an intensifying trend toward outsourcing game development work, with big studios such as EA on a global quest, from Shanghai to Ho Chi Minh, for new sources of skilled game labour. Within even its most privileged echelons, there are no certainties under conditions of globalization.

Contextualizing video game labour: “Our machinery is the mind”

To contextualize our inquiry into game labour, we must briefly outline the structure of the industry and the **main job types game development involves**. The **U.S. \$28 billion global games** business turns on the relation between **developers, who make games, and publishers, who finance and market them** (Lowenstein, 2005). **Publishers’ control over funding, advertising, licensing, and distribution give them enormous power**—a power consolidated in a dozen or so transnational corporations. These include the three console-makers (i.e., Microsoft, Nintendo, and Sony); a cluster of U.S.-based multinationals (e.g., EA, THQ); a similar European-based group (e.g., UbiSoft, Vivendi); and a set of Japanese publishers (e.g., Capcom, Namco). **Many publishers, like EA, operate in-house development studios, though the demand for a steady churn of fresh content means that independent studios, supplying the mega-publishers with games, proliferate**. Concentrated in metropolises like Los Angeles and Vancouver, studios therefore range widely in size, from mammoth 800-person facilities to small ones with a single 20-person team. **In terms of employment figures, the U.S. game industry directly employed some 30,000 people in 2000 (Entertainment Software Association, 2001), while in Canada the number stands at about 6,000 (Dyer-Witthford & Sharman, 2005).**

“Our machinery is the mind,” one studio executive told us (interviewee 2, executive, company name withheld, Vancouver, BC, personal interview, 5 June, 2002). This cerebral machinery is set to work in teams numbering from 20 up to 100, with the **development of a console game costing anywhere from U.S. \$5 million to \$25 million (Geoghegan, 2005) and consuming a period of between 6 and 24 months, depending on a game’s complexity**. The core job categories in game development include designers, artists, programmers, testers, and producers. Designers establish the basic game concept, characters, and play mechanics. Artists develop characters, worlds, textures, animation, and sound. Programmers write the code and create the game engines on which a game’s functionality is based. Testers play a game to evaluate it for bugs and playability. Producers lead the project and manage the development team, trying to maintain a coherent vision of the game’s design and facilitate communication among various subteams.

In terms of work force composition, the game work force is youthful, as it has tended to be since the genesis of the industry in the 1970s. It is now aging slightly, with an **average age of 31, but the largest proportion of game workers is still between their late-teens and thirties** (International Game Developers’ Association [IGDA], 2005). Over 60% of game workers hold college or university degrees,

and a further 16% have graduate degrees (IGDA, 2005). Salaries vary depending on rank, department, experience, and location. Celebrity designers rake in as much as \$400,000 (US). Programmers average some \$70,000, artists about \$60,000 (US), while testers are often on contract and paid minimum wage (IGDA, 2005; Olsen, 2004). **Springing from a Silicon Valley environment outside the orbit of traditional labour movements, game development is a non-unionized sector. There are, however, an active professional association—the International Game Developers Association (IGDA)—and a network of magazines, websites, and industry gatherings, which help connect members of this work force.**

The analysis of video game labour that follows is informed by four main sources. The first is the discussion in online forums, game-industry publications, and the mainstream press generated by EA Spouse's post. The second is a series of studies on "quality of life" in the industry, most notably those produced by the IGDA (2004a; 2004b; 2005). The third is the growing body of academic literature on the video game industry (Kline, Dyer-Witthford, & de Peuter, 2003), including that on its gendered division of labour (Cassell & Jenkins, 1998; Laurel, 2001). Our fourth source is a set of in-depth interviews with game workers we conducted between 2002 and 2004 in the context of a larger study of the political economy of the Canadian video game industry.² Canada is a small but significant node in the global circuits of digital play, and while its game development business has distinctive features (Dyer-Witthford & Sharman, 2005), it is broadly representative of industry norms in North America in terms of workplace conditions.

Enjoyment: "I'm going to work. Cool!"

For millions of young men, and a very few women, to be employed in a game company seems a "dream job" (Chung, 2005). Once hired, many game workers clearly do—or at least at some time did—enjoy their work. **Creativity, co-operativeness, and coolness were among the pleasurable qualities of work in digital play that we were told of in our interviews.** Acknowledging that the scope of creativity enjoyed by game workers of course varies with role and rank, a lead designer conveys a sense of what the industry offers those in its upper reaches:

The best thing is the flexibility and the fact that I can continue to learn new things. It's never really the same. It never gets boring. If I do have a task that is in some way tedious, it's not going to be that way forever. I know that I'm not doing the same thing over and over again. That's absolutely . . . the best thing. (interviewee 3, designer, company name withheld, Vancouver, BC, personal interview, 6 June, 2002)

With responsibilities that include generating original game ideas and working on proofs of concept to pitch to publishers, this designer underscores the degree of artistic freedom he enjoys: "I have a huge amount of **flexibility and autonomy, and a huge amount of support in terms of getting resources for realizing these ideas**" (interviewee 3, designer, company name withheld, Vancouver, BC, personal interview, 6 June, 2002). Similar attractions are found on the programming side, where people who come to game development from computer engineering often express their enjoyment at **"being creative at work and still**

using my technical skills. . . . It's like architecture and engineering rolled into one. . . . " (interviewee 4, programmer, company name withheld, Vancouver, BC, personal interview, 8 August, 2002). Evidence that game developers find their work exciting is not just anecdotal; in response to IGDA surveys, 83% said their job was "constantly or usually stimulating" (2005, p. 17).

Creative freedom is in turn related to workplace organization. In small- and medium-sized development companies in particular, relatively flat structures are commonplace. One software engineer used the concept of "working anarchy" to explain how the game development process unfolds at the mid-sized studio where he works:

There's little bureaucracy. It's just people doing their thing to make good games. . . . We have very little hierarchy, very little formal structure, very little "understood" ways of doing things. . . . In a situation where everyone more or less knows their role everyone just divides the work, you work on your bit, and everyone knows what to do. It just works out. (interviewee 4, programmer, company name withheld, Vancouver, BC, personal interview, 8 August, 2002)

The intensifying consolidation of ownership in the industry is, however, reducing such "working anarchy" in favour of the more rationalized production processes that tend to be found in the giant studios. But even in larger studios, participating in the complex division of labour that making a game entails, and seeing it all weave together, can in itself be exhilarating. One employee described the co-operative aspect of game development as a source of great solidarity and fun:

It's the rush of being involved in a big project. There is a lot of teamwork. You make really good friends, because you all suffer together. You're all up at 4:00 in the morning trying to code and you end up playing a game or watching TV. . . . it's fun. It's almost like being in school and getting paid for it. I went to school and worked really late on team projects, and I really miss that. . . . [T]he game industry provides a little bit of that. (interviewee 6, assistant producer, company name withheld, Vancouver, BC, personal interview, 1 June, 2002)

A third factor that makes game work enjoyable is rooted in these creative and co-operative practices but adds a less tangible layer—an ambience of "cool" built up around unregulated hours, lax dress code, studio pranks, free food, fitness facilities, lavish parties, funky interior design, and an array of other perks and promises. In terms of the cool factor, many of the people we talked to referred to the "rebelliousness" found in game studios, which some contrasted to the "hideous" stiffness and rationality of the "corporate world." "In the industry there are a lot of very bright, very jaded people," one producer explains. "[N]one of our people would ever attend a meeting in a suit" (interviewee 7, producer, company name withheld, Vancouver, BC, personal interview, 5 June, 2002). Such informality is encouraged because managers see it as necessary to inspiring the creativity they must harness: "[L]eeway to express yourself [is essential]. People have to be entirely comfortable to be who they are to come up with anything spontaneously, to have that real dynamic" (interviewee 7, producer, company name withheld, Vancouver, BC, personal interview, 5 June, 2002).

Organizational culture also bends to a work-as-play model because game companies face recruitment and retention problems. Anxious about workers leaving midway through a game project, developers are loath to invest in an employee's training and to provide them with access to contacts in the industry, only for them to go to another company or start their own studio. "Keeping your people is this almost maniacal focus for the people that run companies," explains one studio owner who worked at a number of game development companies before starting his own. "[There is an] almost coercive quality of being manipulated into staying. 'Oh my God, you don't want to leave here!'" (interviewee 1, owner, company name withheld, Vancouver, BC, personal interview, 14 August, 2002). Hip corporate culture also feeds into the off-job cachet of working in the game sector. This benefits the social life of game workers, with one candidly describing the "added bonus" of his job: "If I'm at a bar, and someone cares about it, and I say I'm working [in the game industry], they go, 'Whoo, that's cool.' I try not to perpetuate it but it happens. It's this thing that you can't control" (interviewee 6, assistant producer, company name withheld, Vancouver, BC, personal interview, 14 August, 2002).

Individual creativity, collective co-operation, and a cool aura make for a very attractive package. Indeed, it is impossible to understand the situation of game labour without coming to terms with remarks such as this: "Generally, when you go to work, it's not, 'Ah, I gotta go to work.' It's, 'I'm going to work. Cool!'" (interviewee 8, producer, company name withheld, Edmonton, AB, personal interview, 1 June, 2002). Or: "You come in, you see your friends, you get to make video games, and you get to play some. It's pretty cool. It's really not even so much like work here" (interviewee 9, tester, company name withheld, Vancouver, BC, personal interview, 1 June, 2002). The irony is that the very attractions that make employment in game development "not so much like work" can also forge the iron cage of endless hours described by EA Spouse, converting a dream job into a nightmare.

Exclusion: "It's a total 'old boys' club"

"If you go into a games company," one studio owner warned us, "what you'll basically find is a big room full of all these sort of quiet, essentially anti-social and shy guys, hammering away at their keyboards with their headphones on" (interviewee 1, owner, company name withheld, Vancouver, BC, personal interview, 14 August, 2002). Both our observations and survey-based studies confirm this gender imbalance. As in all supposedly male social domains, there is, however, a female counterhistory to game development (ELSPA, 2004). There have been, and are, female game developers. The mid-1990s saw a "girl game" movement and a wave of entrepreneurial feminist game experiments (Cassell & Jenkins, 1998). Although many of these failed (Laurel, 2001), there is a plausible argument that some of the momentum of this movement was harnessed in the creation of "gender-neutral" games such as *The Sims*, often held up as a flagship of gender equity (Jenkins, 2003). There is also a tradition of feminist game "modders" challenging norms and representation of gender in game culture (Schleiner, 2005).

Nonetheless, game development is an overwhelmingly masculine dungeon. An estimate we received from a female insider was that **women account for** an average of only **10-15% of a game developer's staff**. An IGDA (2005) survey, which garnered some 4,000 responses, mostly from North America, found that women made up only **11.5% of respondents**. Of these, **few are in senior positions**. IGDA found that "male workers heavily dominate most of the core content creation roles" (2005, p. 12) and that a \$9,000 (US) annual compensation gap exists between women and men in comparable positions. Despite marginal shifts in the late 1990s, the verdict of most women we interviewed on the industry's gender balance was scathing. Said one, "It's a total 'old boys' club" (interviewee 11, details withheld on request, personal interview, 2002).

This exclusion of women can be traced to gaming's genesis, and it is perpetuated and reinforced at multiple levels. First, many developers hire candidates either with previous experience in the games industry or at least with a passion for play. Since **historically games have been marketed primarily to a "testosterone niche"** (Kline et al., 2003, p. 257), **the result has been an industry built around games made by males for males**. Secondly, there is the gendered nature of many of the disciplines that provide training in requisite game development skills—for example, computer science, where female enrolment is actually declining in North America. Thirdly, there is **risk aversion in the game industry**, which has historically been reflected in a **lack of funding for start-up developers who want to experiment with a game that might appeal to a cross-gender audience**:

If you were to ask any of the women who have tried . . . they do not get support in the industry at all. . . . that's the problem when you have a really dominant gender leading and they're the ones who have the purse strings. (interviewee 11, details withheld on request, personal interview, details withheld on request, 2002)

That game play is de facto training for game work generates an obvious circularity in work force composition. While some men we interviewed expressed a desire for greater gender balance, they noted a serious obstacle in the fact that women played less than men: "[W]e see it is a problem and we'd like to see it change, but feel like there's not too much that we can do about it right now" (interviewee 4, programmer, company name withheld, Vancouver, BC, personal interview, 8 August, 2002). Several team leaders stated as a regrettable but unavoidable fact of life that while male and female job applicants might have the same qualifications on paper, the former nearly always had more gaming experience. Others offered a clearer window into the sexism of the industry, explaining that "girls" often do not have "the right ideas" when it comes to games (interviewee 12, producer, company name withheld, Ottawa, ON, personal interview, 8 June, 2002) or that the main reason for hiring women is that it "looks good" for a developer to employ "some girls" (interviewee 13, artist, company name withheld, Vancouver, BC, personal interview, 8 June, 2002).

Undoubtedly, games such as The Sims have in the last decade attracted female players, and casual online gaming is a field where many women play. But

some recent examinations of digital culture suggest that "visions of inclusive gaming . . . remain some way off" (BBC News, 2005). Female game workers themselves point out that there remains "huge risk aversion" in this sector, with publishers and developers concentrating on males as the "core demographic." Even if there has been a shift in the gender of game players, "there's not much of a change in hiring numbers," we were told (interviewee 7, producer, company name withheld, Vancouver, BC, personal interview, 5 June, 2002). As one analyst observes: "An already male dominated workforce and tribes of determined male hardcore gamers means that unless companies make an effort to look at less traditional sources of recruitment, they may never see many female applicants" (Haines, 2004, p. 6).

As a masculine dungeon, the game studio is a place of creative camaraderie, technological intensity, and cerebral whimsy, but it is also often obsessively hard driving, punishingly disassociated from rhythms of domesticity, sleep, and nourishment. In yet another feedback loop, however, the insane hours of work (addressed in the next section) extracted from this male-dominated cultural activity and workplace in turn become a barrier to the participation of women, who will often carry the burden of a "second shift" (Hochschild, 1990) of child-care and domesticity awaiting them at home. The construction of the game studio as "a world where women are referred to as 'ladies,' where to go home early is to let the team down, and to fit in is to be seen as 'laddish'" (Haines, 2004, p. 7) means that the contribution of women to game development is often invisible, taking the form of the unacknowledged labour charted by a generation of feminist scholars (Dalla Costa & James, 1972; Waring, 1990)—in the domestic activities of the mothers and partners who clean up after, take care of children for, and provide emotional sustenance to a predominantly male work force. This brings us back to the exploitive conditions exposed by EA Spouse.

Exploitation: "You can sleep here all night"

While there are several sources of discontent in game studios, the length of the working day—and night—is the secret whose disclosure by EA Spouse has deeply embarrassed video game corporations and galvanized many of their workers. The length of the working day varies widely, depending on the company, the stage a team is at in the development process, a worker's role on a project, and the worker's slot in the hierarchy. But the IGDA survey reports that almost 60% work 46 hours or more a week, and nearly 20% work over 55 hours. These hours are subject to regular intensifications, with one interviewee telling us digital play is an industry where "circadian rhythm is regularly broken" (interviewee 14, artist, company name withheld, Montréal, QC, personal interview, 15 August, 2004). "Crunch time" is the industry term that indicates an apparently unusual period of crisis in the production schedule. During crunch time, more than 35% of respondents reported working 65-80 hours a week, with 13% doing more than 80 hrs; reports of working more than 100-hour weeks are not unheard-of (IGDA, 2004a). Perhaps the most startling thing about the IGDA report is that more than half of respondents said that "management sees crunch as a normal part

of doing business,” with only about 2% actively implementing “no crunch policies” (IGDA, 2004a, p. 19). Crunch time has become “built into the equation” (Hyman, 2005).

Why do these situations arise? Under the best of circumstances, maintaining an orderly workflow is an organizational challenge for studios. As IGDA notes, “significant chunks” of the workflow are often outside the developer’s control, as publishers may demand design changes, licensed assets may be delayed, or third-party tools and libraries may be “late, buggy, or both” (2004a, p. 13). On top of this, game developers work to unforgiving deadlines. Pressure in this direction arises from the revenue model that keeps studios afloat. A publishing contract identifies a series of scheduled milestones; each time the developer meets one, a payment is triggered. As one programmer said, “There’s a lot of pressure when you’re looking at a deadline and something just has to work and it doesn’t, but you just have to make it work by next Tuesday. You end up working really long hours” (interviewee 4, programmer, company name withheld, Vancouver, BC, personal interview, 8 August, 2002). While developers with a hit game behind them may be able to negotiate tolerable deadlines, vulnerable start-ups often complete deals that place intense demands on workers. “Sometimes companies are just so intent on getting that contract that they’ll promise anything . . .” (interviewee 13, artist, company name withheld, Vancouver, BC, personal interview, 8 July, 2002). And as a game development project approaches conclusion, things become more frantic, as “it is often true that the ship date is impossible to delay without catastrophic consequences like cancellation or even bankruptcy” (IGDA, 2004a, p. 13). While the major studios do not suffer exactly the same time pressures as small studios, they are exposed to others, such as the need to complete a game for the all-important Christmas season, to synchronize with the start of a sport’s season, or simply to clear the decks for the next game in a relentless stream of projects.

A point often made in the discussions around EA Spouse is that the “garage invention” model that lies in the roots of the game industry is not well fitted to meet these challenges. From this point of view, the “working anarchy” of small studios, while perhaps favouring creativity, is all too likely to fail at crisis avoidance. This is especially the case, it is argued, as games demand larger teams. IGDA’s Della Rocca observes that when game production involves “200 plus people and a production budget of maybe \$20 million, that garage band approach to things doesn’t scale to match” (cited in Chung, 2005, p. R7). In this logic, the overwork issue is a problem of industry “maturity,” a failure to develop sufficient managerial skills and organizational competence to keep pace with industry success, and, by implication, a problem that can be dealt with by a process of education.

There is some validity to this explanation. But it has one obvious weakness. If recurrent crunch time results from insufficient managerial experience, one would expect the worst offenders to be new, small companies. And there is no shortage of horror stories from such places. But EA Spouse’s complaint deals with a well-

established studio: EA has been making games since 1982. Many of these games are among the most formulaic and hence plan-able products in the business. Its sports franchises, such as Madden NFL and FIFA—the sort EA Spouse's partner was working on—are infamous as the type of unoriginal games that are updated annually with the addition of some new features and statistics. If any company could be expected to overcome the managerial problems of preventing overwork, it would be EA. EA Spouse, speaking of the "crunch" in which her partner suffered, wrote:

Every step of the way, the project remained on schedule. Crunching neither accelerated this nor slowed it down; its effect on the actual product was not measurable. The extended hours were deliberate and planned; the management knew what it was doing as it did it. (EA Spouse, 2004)

Normalized crunch time therefore points to a very elementary economic fact: it is a good deal, a steal in fact, for game companies. The IGDA survey (2004a) showed that for just under half of respondents, overtime was uncompensated. When it is compensated, it usually is not in the form of direct payment. The most common forms of compensation are time off at the end of the project, royalties, and profit-sharing; only 4% of respondents said their companies paid overtime in cash.

An important factor in all of this is the labour legislation that enables video game companies to exempt high-tech workers from overtime payment. In the U.S., the Fair Labor Standards Act (FLSA) exempts companies from paying overtime to computer professionals engaged in a strictly defined set of tasks and making over a certain amount per hour. This has often been interpreted by companies as an excuse to withhold such payments from all salaried programmers. However, each state has its own labour regulations. In the event that the FLSA and any state laws differ, the employer must follow the law or rule that provides the greatest protection to the employee. Labour law in California, where EA and other major publishers have studios, stipulates companies do not have to pay overtime to software programmers if they make more than \$41 (US) an hour and engage in advanced work that is creative or intellectual in nature. In Canada, British Columbia, Alberta, and Ontario also have overtime exclusions for high-tech workers, and, in British Columbia, EA and other game companies lobbied vigorously to secure this deregulation.

Why have game workers put up with these long hours? This is, at first sight, puzzling. Demand for skilled programmers and designers is high. Companies anxious about losing talent would seem to have an incentive to treat well workers who could quit to join a competitor. In part, the answer to this puzzle is that while experienced game workers are in short supply, new entrants are plentiful. Though excessive hours are widespread, they are disproportionately endured by the youthful contingent of the work force. "[S]o many people in the video game industry are like nineteen or twenty—just fresh out of school," says one game artist who recalls his own first game job. "I had never even had a dishwashing job before. I was working fourteen-hour days and never seeing the light of day, which

was great at the time. I just stayed there all the time.” Consenting to such hours, he says, “partly has to do with the fact that they promote, you know, ‘Hey, we have a couch here. You can sleep here all night.’ . . . You’re nineteen” (interviewee 13, artist, company name withheld, Vancouver, BC, personal interview, 8 July, 2002).

One studio owner, who had also worked at other developers, was equally straightforward:

Companies tend to get these young guys that come out of film school, game programming school, or art school and get them to work their asses off. The mechanism for doing that is the game industry’s corporate culture: “You don’t have to leave because we give you all the Pepsi and all the potato chips you’d ever want.” (interviewee 1, owner, company name withheld, Vancouver, BC, personal interview, 14 August, 2002)

The stamina of youthful game workers helps set a studio norm of overwork, an instance of what Angela McRobbie refers to as “enforced youthfulness” (2002, p. 110), so prominent in many creative-industry workplaces.

Excessive hours are further reinforced by a variety of incentives. While smaller studios can offer chips and a couch to sleep on, the attractions proffered by larger ones are more extravagant. One of the most striking examples is EA’s massive campus-like Vancouver studio. Specializing in the sport genre, EA Canada’s complex employs about 1,000 people and features a fully equipped gym, pool tables, basketball courts, a soccer field, and snowboarding field trips. Even the former president of EA Canada, Glenn Wong, admits his company’s spectacular facility is “just candy”: “Here it is, 3:30, a gorgeous afternoon, and my soccer field is empty. But I can tell you that at 3:30 this morning, there will be 75 people in this building working their butts off” (cited in Taylor, 1999 n.p.). Why? “The guts of it that makes it a cool place to be is that the people here want to win. Trying is nice, making mistakes is okay, but it’s all about winning.”

Even after the appeal of the perks wanes, other corporate strategies bind workers to the workplace. One of the common forms of compensation for unpaid overtime is what one developer calls “the Golden Shackles”:

You work on a game and they offer you a profit-sharing agreement, but you have to stay at the company to take advantage of it. So you work for two years on a game with the intent that if it sells a lot then you’ll get a share of that, and then it takes another six months to get the game to market, and then it takes another six months before the money starts to filter back. So you’ve got this employee who stuck around for at least another year to get in on that profit sharing and, by this time, they’ve already started on another game and are sort of stuck there. (interviewee 1, owner, company name withheld, Vancouver, BC, personal interview, 14 August, 2002)

At the start of one’s career, another worker admits, stock options can be “totally compelling. People at first are saying, ‘Holy shit, I’m getting stock options.’” But eventually the gold on the shackles flakes: “Take your stinking stock options away from me! I’m happy having time to myself. I need time to recharge. This isn’t my life. It’s a part of my life, but not the whole thing” (inter-

viewee 13, artist, company name withheld, Vancouver, BC, personal interview, 8 July, 2002).

A further factor is the volatility of the industry. Since the labour market is tight, a game worker unhappy with an established company like EA might well be able to get a job somewhere else. But, given the high rate of bankruptcies in the business, he would also have to calculate the possibility that his new employer would vanish within a year or so. Or, indeed, be bought up by EA or some other giant publisher. This was a point raised by EA Spouse, who cites the "collapse of dozens of small game studios, no longer able to acquire contracts in the face of rapid and massive consolidation of game publishing companies," as a reason why EA could get away with its alleged "If they don't like it, they can work some place else" policy (2004).

Many of the enjoyments of game development that we described earlier can become links in the chains of overwork. Passion to make a good product, team solidarity, and social glamour are all part of the allure of game labour. But these also produce a hard-driving, largely internalized work ethic intensely beneficial to the employer's bottom line. To this is added the "concept of ownership," actively deployed by some studios:

When you are responsible for something in a game, you "own" it. If something goes wrong with that [part of the] game after release, you can pretty much kiss your ass goodbye That's where a lot of the stress comes from. [Y]ou're not supposed to do overtime, but you don't mind doing it because you're given "ownership" When you work in this industry . . . you are judged for what you've done. So you want to make a good name for yourself. You want people to consider you a hard worker, a good worker; a guy that can do a bit more than what's expected, because the thing with the game industry is that it is, really, a small business. A lot of guys have made the mistake of quitting work during "alpha" or something. They'll never get a job in the industry again. (interviewee 6, assistant producer, company name withheld, Vancouver, BC, personal interview, 1 June, 2002)

Youthful enthusiasm, home-away-from-home workplaces, stock options, the risks of leaving, macho bravado, and a cool corporate culture—these are among the softly coercive elements of video game companies' culture of extreme work. Indeed, the IGDA speaks of a culture of "forced workaholism" (2004a, p. 6). While acknowledging that there are game companies that pursue responsible and humane management strategies, its aggregate portrait is one of "horrible working conditions" (IGDA, 2004b, p.1).

Along with a severe work regimen, workers highlight stress and health issues. Asked how they felt after extended periods of crunch time, the responses of workers interviewed by the IGDA ranged from "exhausted" to "flipped out" (2004a, p.71). Referring to the systemic extreme hours, one artist told us, "I don't think it's good for you to work like that that often, and to be creative all the time without a break. It just isn't good for your brain." Stress can be huge. "[At some studios] where you're expected to work those hours . . . or you've done all this work and there is no recognition—you feel like crap afterward. It's ongoing.

‘Holy shit, I’ve totally been screwed around’” (interviewee 13, artist, company name withheld, Vancouver, BC, personal interview, 8 July, 2002).

Some of the most tragic aspects of the picture emerging in the wake of EA Spouse’s disclosures are the testimonials about social and domestic relationships. A programmer from a Canadian studio speaking of “the havoc the game industry wreaked in my personal life” says his “last 3 girlfriends blamed the game industry as the reason our relationship didn’t work out” (IGDA, 2004a, p. 76). A game tester for a U.S. company remarked, “[A] lot of people become alcoholics, they lose their families, they miss their kids,” because they are “just at work all the time” (IGDA, 2004a, p. 81). Disgruntled workers refer to studios such as EA as “divorce factories” (cited in Takahashi, 2005a). A game designer for a studio in St. Catharines, Ontario, who on one occasion worked 45 hours straight and on another did 120 hours in a single week, “almost got fired for taking Saturday off to celebrate his wedding anniversary” and finally quit when he and his wife were expecting their first child: “If I had stayed, I probably wouldn’t have been able to go to the hospital when my wife was in labour” (cited in Chung, 2005, p. R5).

Exodus: “The only thing that will get publishers to budge is unionization”

It is hardly surprising that many game workers seek a way out from the conditions we have described. IGDA (2004a) discovered an exceptionally high rate of turnover in the industry, with a growing number of game developers leaving the sector altogether: more than 50% plan to leave the industry within ten years, 35% within five years. Other workers are exploring varied lines of fight or flight. Discontent in the game industry has for some time expressed itself, for example, in clandestine activities such as leaking a game online before its official release date (de Peuter & Dyer-Witthof, 2005). Below we review some more institutional options game workers are pursuing, including legal challenges, educational approaches, union organizing, and entrepreneurial escape.

Around the time EA Spouse’s blog post appeared, several groups of game development workers were launching class-action suits against their employers. One, *Kirschenbaum v. Electronic Arts*, filed in California, alleged EA improperly classified some of its employees so as to avoid paying them overtime (Feldman, 2004). Kirschenbaum’s lawyers argued their client’s job as an “image production employee” did not entail original, creative work, and as a result, he and others in similar positions are eligible for overtime pay. California’s overtime exemption does not apply to image-effects workers in the entertainment industry, of which, they argued, EA is part (Takahashi, 2005a). In 2005 the case was settled out of court, costing EA U.S. \$15.6 million. The settlement, which specifies that future entry-level EA employees will not receive stock options but will be eligible for overtime pay, has been hailed as marking a revolution in Silicon Valley culture. A series of similar class-action suits remain unresolved.

Meanwhile debates about remedies for the labour crisis rage. While there is wide agreement about the need to slow the production process and reduce the hours game workers surrender to work, two different approaches are appearing. One advocates an educational strategy designed to enlighten management: “We

need to educate the middle managers, the project managers, and the producers—or bring in outside management to deal with the chaos and the fires and the pressures of managing large-scale, big budget projects" (Della Roca, cited in Hyman, 2005). An outcome of this conciliatory approach has been an extensive set of forums and papers on "best practices," designed to minimize the situations that provide the official excuse for crunch time (Howie, 2005). One welcome aspect of these discussions about "work-life" balance is that they are increasingly linked to those about gender equity in the workplace, with the "long hours culture" seen as both a cause and effect of the industry's institutionalized sexism (Haines, 2004, p. 13).

The other approach argues that the appropriate response is unionization. Della Roca describes IGDA as "union neutral" (cited in Hyman, 2005), and, within its forums, opinions fall strongly on either side of the issue. Some see parallels between the tumult in the game industry of the 2000s and that in Hollywood in the 1920s and 1930s, the latter resulting in the emergence of the Screen Actors Guild and Writers Guild of America. Some insist big publishers are not going to "benevolently change today's abysmal work conditions without pressure. They will make small changes, but not much else, if the threat of unionization seems real" (T. McPherson cited in Hyman, 2005). Other game workers are looking to labour organizations such as WashTech (Washington Alliance of Technology Workers), a local of the Communication Workers of America—a cross-sectoral union that has started some organizing efforts among game workers. EA Spouse has come out in favour of unionization, observing that while the spate of publicity about work hours has temporarily curbed the imposition of permanent crunch time, "I don't think that will be very long-lived. In my opinion, the only thing that will get publishers to budge is unionization, which I believe to be the best solution" (cited in Hyman, 2005).

Union organizing will, however, face major obstacles. The strong entrepreneurial ethos of game development would seem to be antithetical to traditional labour-movement culture. Many game development workers themselves tolerate bad working conditions because they see a period of corporate drudgery as a step to starting their own companies. A symptom of the labour crisis in the industry, and of an overall creativity crisis arising from publishers' dependence on clones and franchises, has been a revived interest in indie games, expressed in initiatives such as Manifesto Games. These projects express the aspiration of game developers to increase their control over the quality and content of their work, an ambition they aim to fulfill by constructing small companies committed to realizing the creative potential of games. Such attempts to diversify video game culture deserve support. But we suspect this path of entrepreneurial exodus from the labour crisis will be difficult in the face of the current wave of corporate consolidation.

The other barrier to collective organization by game workers is, of course, the reactions of game companies to the rumblings of dissent. EA Spouse's posting, coupled with a string of lawsuits, has shocked game corporations, throwing them

on the defensive. One result has been a flood of promises to improve working conditions. UbiSoft's Montréal studio appointed a "VP of continuous improvement" to address quality-of-life and workflow issues and created a 60-person "bureau de project" dedicated to "planning and streamlining production," with one aim being to reduce crunch time (Chung, 2005, p. R7). At the same time, some corporations have asserted that long hours arise "more from a certain bravado or peer pressure than from management" (Allard, cited in Hyman, 2005). In EA's proposals for reform, the desire to prevent unionization is unambiguous. While claiming that EA is "in the forefront" of addressing "work-life balance," the developer's Human Resources VP warns against "people who want to step in and take a piece of the pie or get in the middle of things without contributing to the growth of the business" (cited in Hyman, 2005).

Many are sceptical about the sudden flurry of corporate good intentions. One of the lawyers representing aggrieved workers in the above-mentioned Kirschenbaum case suggested "most employers rely on their employees being hesitant to bring lawsuits and just hope it will blow over" (Graves, cited in Chung, 2005, p. R7). There were reports that in the wake of lawsuits EA had decided to "move hundreds of employees to Florida and Canada after being forced to reclassify which positions are eligible for overtime in California" (Feldman, 2005). This type of strategy—a corporate riposte of fight and flight—is likely to become much wider in scope. Offshore outsourcing is beginning to hit the gaming sector. EA this year opened a centre in Singapore. And EA's recent appointment to its Board of Vivek Paul, vice-chairman at Wipro, one of the leading companies in providing software outsourcing to India, is seen as a sign that it may be looking toward centres on the subcontinent to find a cheaper labour force (Takahashi, 2005b). The huge fixed investment represented by game studios in places like Vancouver, Montréal, and California will probably ensure that in the near- to mid-future much of the high-concept game development remains at these locations, even if formulaic components are increasingly outsourced. In the longer term, the North American and European game-labour force will have to wage their fight for survivable working hours across a global battlefield.

Conclusion: "Everyone works too much"

Despite the difficulties that organizing videogame workers will face, the recent revelations of the industry's labour problems constitute an advance. The video game industry's work-as-play ethos has been one small element in an overarching mythology that presents digitization as dissolving the contradictions and conflicts of capitalism. The shattering of this ethos is a step toward a more realistic assessment of digitized work conditions. One could see the story told by EA Spouse as just a disclosure of the problems arising from a very specific industry, with an unusual history, an extreme gender bias, and a unique corporate culture. To a degree, this is true. But for all its exceptionalism, the conditions of the video game industry are suggestive of a broader tendency. Indeed, one of the strengths of the IGDA "Quality of Life" white paper is that it opens its examination of long studio hours by observing that while the problems it documents may be "particularly

strident in the game industry, we do not hold a monopoly on them by any stretch of the imagination" (2004a, p. 10), and it substantiates this observation with a section headed "Everyone Works Too Much," which places these issues in the context of a broader, and well-documented, North American crisis of workplace stress (Menzies, 2005; Schor, 1993).

From this perspective, anyone inclined to read our paper only as an account of the workplace troubles into which an echelon of young male game workers with a dubious cultural obsession have fallen might reflect on how similar their problems of long hours, boundary-less toil, and workplace burnout are to those suffered by an apparently very different group of workers—academics. The implications of this paper's story of overwork with respect to strategies of organized labour are also suggestive. For the conclusions drawn by EA Spouse are similar to some made more than a century ago—namely, that if one wishes for a life in which human energy can "blossom forth," then "the shortening of the working-day is its basic prerequisite" (Marx, 1981, p. 959). To strategize in this direction would be to take seriously, with EA Spouse, one of EA's corporate mottos: "Challenge Everything!"

Notes

1. The identity of EA Spouse was disclosed on April 26, 2006. Erin Hoffman is married to Leander Hasty, an engineer at EA in Los Angeles. Hasty helped launch a now-settled class-action lawsuit against EA for unpaid overtime.
2. This research was supported by Canada's Social Sciences and Humanities Research Council. We want to express our gratitude to the developers whom we interviewed for this study. Because our interviews dealt with volatile issues in a tight-knit business with little job security, this paper preserves the anonymity of our interviewees.

References

- BBC News. (2006, January 9). *Games firms face challenge ahead*. URL: <http://news.bbc.co.uk/2/hi/technology/4587902.stm> [April 25, 2006].
- Barbrook, Richard, & Cameron, Andy. (1996). The Californian ideology. *Science as Culture*, 6(1), 44-72.
- Cassell, Justine, & Jenkins, Henry. (Eds.). (1998). *From Barbie to Mortal Kombat: Gender and computer games*. Cambridge, MA: MIT Press.
- Chung, Emily. (2005, August 15). Dream jobs in hell. *The Toronto Star*, p. R5-7.
- Dalla Costa, Mariarosa, & James, Selma. (1972). *The power of women and the subversion of the community*. Bristol, UK: Falling Wall Press.
- de Peuter, Greig, & Dyer-Witthorff, Nick. (2005). A playful multitude? Mobilising and counter-mobilising immaterial game labour. *Fibreculture Journal*, 5. URL: http://journal.fibreculture.org/issue5/depeuter_dyerwitthorff.html [September 13, 2006].
- Dyer-Witthorff, Nick. (1999). The work in digital play: Video gaming's transnational and gendered division of labour. *Journal of International Communication*, 6(1), 69-93.
- Dyer-Witthorff, Nick. (2001). Nintendo capitalism: Enclosures and insurgencies, virtual and terrestrial. *Canadian Journal of Development Studies*, 22, 965-997.
- Dyer-Witthorff, Nick. (2005). Cyber-Negri: General intellect and immaterial labour. In T. Murphy & A.-K. Mustapha (Eds.), *The philosophy of Antonio Negri: Resistance in practice* (pp. 136-162). London, UK: Pluto.

- Dyer-Witheford, Nick, & Sharman, Zena. (2005). The political economy of Canada's video and computer game industry. *Canadian Journal of Communication*, 30(2), 187-210.
- EA Spouse. (2004, November 10). *EA: The human story*. URL: <http://ea-spouse.livejournal.com/274.html>.
- Entertainment & Leisure Software Publishers Association (ELSPA). (2004). *Chicks and joysticks: An exploration of women and gaming*. URL: http://www.elspa.com/assets/files/c/chicksandjoysticksanexplorationofwomenandgaming_176.pdf [September 13, 2006].
- Entertainment Software Association (ESA). (2001). *Economic impacts of the demand for playing interactive entertainment software*. URL: http://www.nathaninc.com/NATHAN/files/ccPageContentdocfile_name000553705546idsareport.pdf [September 13, 2006].
- Feldman, Curt. (2004, November 11). Employees readying class-action lawsuit against EA. *Gamespot*. URL: http://www.gamespot.com/news/2004/11/11/news_6112998.html [April 27, 2006].
- Feldman, Curt. (2005, October 5). EA settles labor-dispute lawsuit. *Gamespot*. URL: http://www.gamespot.com/news/2005/10/05/news_6135106.html [April 27, 2006].
- Geoghegan, John. (2005, December 14). The console transition: A publisher's perspective. *Business Week*. URL: http://www.businessweek.com/print/innovate/content/dec2005/id20051214_971333.htm [April 27, 2006].
- Haines, Lizzie. (2004). *Why are there so few women in games?* URL: http://www.igda.org/women/MTNW_Women-in-Games_Sep04.pdf [April 27, 2006].
- Hardt, Michael, & Negri, Antonio. (2000). *Empire*. Cambridge, MA: Harvard University Press.
- Hardt, Michael, & Negri, Antonio. (2004). *Multitude: War and democracy in the age of empire*. New York, NY: The Penguin Press.
- Hochschild, Arlie. (1990). *The second shift*. New York, NY: Avon.
- Howie, Hank. (2005, January 31). Making great games in 40 hours per week. *Gamasutra*. URL: https://www.gamasutra.com/php-bin/login.php?from=http://www.gamasutra.com/features/20050131/howie_01.shtml [September 13, 2006].
- Hyman, Paul. (2005, March 22). Unionize now? *Gamasutra*. URL: http://www.gamasutra.com/features/20050322/hyman_01.shtml [September 13, 2006].
- International Game Developers' Association (IGDA). (2004a). *Quality of life in the game industry: Challenges and best practices*. URL: <http://www.igda.org/qol/white-paper.php> [September 13, 2006].
- International Game Developers' Association (IGDA). (2004b, November 16). *Quality of life issues are holding back the game industry: Open letter from IGDA Board of Directors*. URL: http://www.igda.org/qol/open_letter.php [September 13, 2006].
- International Game Developers' Association (IGDA). (2005, October 5). *Game developer demographics: An exploration of workforce diversity*. URL: http://www.igda.org/diversity/IGDA_DeveloperDemographics_Oct05.pdf#search=%22Game%20developer%20Demographics%22 [September 13, 2006].
- Jenkins, David. (2004, July 1). Programmer sues Vivendi over long hours, *Gamasutra*. URL: http://gamasutra.com/php-bin/news_index_plain.php?story=3971 [April 25, 2006].
- Jenkins, Henry. (2003). From Barbie to Mortal Kombat: Further reflections. In A. Everett & J. T. Caldwell (Eds.), *New media: Theories and practices of digitextuality* (pp. 243-254). New York, NY: Routledge.

- Kline, Stephen, Dyer-Witthford, Nick, & de Peuter, Greig. (2003). *Digital play: The interaction of technology, culture, and marketing*. Montréal, QC & Kingston, ON: McGill-Queen's University Press.
- Laurel, Brenda. (2001). *Utopian entrepreneur*. Cambridge, MA: The MIT Press.
- Lowenstein, Douglas. (2005, May 18). *E3 Expo 2005 state of the industry address*. URL: http://www.theesa.com/archives/2005/05/e3expo_2005_sta.php [April 27, 2006].
- Lugo, Jairo, Sampson, Tony & Lossada, Merlyn. (2002). Latin America's new cultural industries still play old games. *Game Studies*, 2(2). URL: <http://www.gamestudies.org/0202/lugo/> [April 27, 2006].
- Marx, Karl. (1981). *Capital: A critique of political economy, Volume 3*. New York, NY: Vintage. (Originally published in 1894)
- McRobbie, Angela. (2002). From Holloway to Hollywood: Happiness at work in the new cultural economy? In P. du Gay & M. Pryke (Eds.), *Cultural economy* (pp. 197-214). London, UK: Sage.
- Menzies, Heather. (2005). *No time: Stress and the crisis of modern life*. Toronto, ON: Douglas & McIntyre.
- Mosco, Vincent. (2004). *The digital sublime: Myth, power, and cyberspace*. Cambridge, MA: The MIT Press.
- Olsen, Jennifer. (2004, February 11). Game developer's salary survey. *Gamasutra*. URL: http://www.gamasutra.com/features/20040211/survey_pfv.htm [September 13, 2006].
- Ross, Andrew. (2003). *No-collar: The humane workplace and its hidden costs*. New York, NY: Basic Books.
- Schleiner, Anne-Marie. (2005). Game reconstruction workshop: Demolishing and evolving PC games and gamer culture. In J. Raessens & J. Goldstein (Eds.), *Handbook of computer game studies* (pp. 405-414). Cambridge, MA: The MIT Press.
- Schor, Juliet. (1993). *The overworked American: The unexpected decline of leisure*. Boston, MA: Basic Books.
- Takahashi, Dean. (2005a, April 4). Video-game workers sue for overtime pay. *San Jose Mercury News*. URL: <http://www.technewsworld.com/story/38451.html> [April 27, 2006].
- Takahashi, Dean. (2005b, July 05). India's game developers target outsourcing. *San Jose Mercury News*. URL: http://blogs.mercurynews.com/aei/2005/07/profile_indias_.html [September 13, 2006].
- Taylor, Chris. (1999, November). Best companies to work for in BC. *BC Business Magazine*, 27(11), 31-51.
- Terranova, Tiziana. (2004). *Network culture: Politics for the information age*. London, UK: Pluto.
- Tsianos, Vassilis, & Papadopoulos, Dimitris. (2006). *Precarity: A savage journey to the heart of embodied capitalism*. URL: <http://www.geocities.com/immateriallabour/tsianospapadopaper2006.html>.
- Waring, Marilyn. (1990). *If women counted: A new feminist economics*. San Francisco, CA: Harper.
- Wright, Steve. (2006). Reality check: Are we living in an immaterial world? *Mute*, 2(1), 34-45.